

Fundamentals of Calculus and Quadratic Equations

1. The Quadratic Formula and Discriminant

In elementary algebra, the quadratic equation is given in standard form as:

$$ax^2 + bx + c = 0 \text{ (where } a \neq 0 \text{)}$$

The solutions to this equation can be found using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The expression inside the square root, $\Delta = b^2 - 4ac$, is known as the discriminant.

- If $\Delta > 0$, there are two distinct real roots.
- If $\Delta = 0$, there is exactly one repeated real root ($x = -b / (2a)$).
- If $\Delta < 0$, there are two distinct complex conjugate roots.

Example 1: Finding Roots

Solve for x in the equation: $2x^2 - 4x - 6 = 0$.

Here, $a = 2$, $b = -4$, $c = -6$.

$$\text{Discriminant } \Delta = (-4)^2 - 4(2)(-6) = 16 + 48 = 64.$$

Since $\Delta > 0$, there are two real roots:

$$x = \frac{4 \pm \sqrt{64}}{2 \cdot 2} = \frac{4 \pm 8}{4}$$

$$x_1 = \frac{4 + 8}{4} = 3$$

$$x_2 = \frac{4 - 8}{4} = -1.$$

Therefore, the roots are $x = 3$ and $x = -1$.

2. Basic Differentiation Rules

Calculus focuses on rates of change and accumulation.

2.1 The Power Rule

For any real power n , the derivative of $f(x) = x^n$ is:

$$f'(x) = \frac{d}{dx} [x^n] = n \cdot x^{(n-1)}$$

2.2 The Product Rule

If $u(x)$ and $v(x)$ are differentiable functions, then the derivative of their product is:

$$\frac{d}{dx} [u(x) \cdot v(x)] = u'(x) \cdot v(x) + u(x) \cdot v'(x)$$

Example 2: Derivative Calculation

Find the derivative of $f(x) = 3x^4 - 5x^2 + 7$.

Applying the power rule term by term:

$$f'(x) = 3 \cdot (4x^3) - 5 \cdot (2x^1) + 0$$

$$f'(x) = 12x^3 - 10x.$$

3. Integration by Substitution

Integration is the reverse process of differentiation. When evaluating complex integrals, u-substitution is often used.

Given the integral:

Integral of $f(g(x)) * g'(x) dx$

Let $u = g(x)$, then $du = g'(x) dx$.

The integral becomes:

Integral of $f(u) du$

Example 3: Definite Integral

Evaluate the integral of $2x * (x^2 + 1)^3 dx$ from $x = 0$ to $x = 1$.

Let $u = x^2 + 1$. Then $du = 2x dx$.

When $x = 0$, $u = 0^2 + 1 = 1$.

When $x = 1$, $u = 1^2 + 1 = 2$.

The integral transforms to:

Integral from 1 to 2 of $u^3 du = [u^4 / 4]$ from 1 to 2

$= (2^4 / 4) - (1^4 / 4)$

$= (16 / 4) - (1 / 4)$

$= 4 - 0.25 = 3.75$ (or $15/4$).

4. Trigonometric Identities

Fundamental trigonometric identities essential for problem-solving:

1. Pythagorean identity: $\sin^2(\theta) + \cos^2(\theta) = 1$

2. Double angle formula for sine: $\sin(2\theta) = 2 * \sin(\theta) * \cos(\theta)$

3. Double angle formula for cosine: $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = 2*\cos^2(\theta) - 1 = 1 - 2*\sin^2(\theta)$

Example 4: Trigonometric Evaluation

If $\sin(\theta) = 3/5$ and θ is in quadrant I, calculate $\cos(2\theta)$.

Using $\cos(2\theta) = 1 - 2*\sin^2(\theta)$:

$\cos(2\theta) = 1 - 2 * (3/5)^2$

$\cos(2\theta) = 1 - 2 * (9/25) = 1 - 18/25 = 7/25$.